Heliopolis-AI

Solar Project Intelligence Agent

Engineering-grade, neural-symbolic due diligence for LSS & C&I solar in Malaysia

Track T1: AI for Clean Energy (Solarvest Holdings Berhad)

Multimedia University (MMU), Melaka, Malaysia

github.com/lewbei/heliopolis-ai

Solar developers lose 2–6 weeks on every site intake

- Land bottleneck: Geran scans use Ekar/Rood/Pole, fractured co-owners, hidden Caveat/Charge/Malay-Reserve flags.
- Grid bottleneck: SLDs must satisfy TNB ESAG / MS 1837:2010; errors trigger expensive rejections.
- Terrain risk: Satellite photos hide >15° slopes and monsoon landslide hazards.
- Financial uncertainty: LSS PPA, C&I NEM, and self-consumption tariffs produce radically different NPVs.

Heliopolis-AI automates land, grid, terrain & finance

Scanned Deed / SLD

Gemini Vision OCR

Legal Parser → Civil Slope → SLD Graph → Financial Engine

SEDA + TNB Prefilled Manifests

All math is deterministic — no LLM hallucination on engineering or finance.

Demo output: C&I NEM / ATAP scenario

Site	**Johor, 9.3 ha**
DC Capacity	**6.90 MWp**
Annual Yield	**~10,600 MWh/year**
Final CAPEX	**RM 20.7M**
NPV (21 yrs)	**RM 10.5M**
IRR	**12.38%**
Land Legal Status	**Flagged/Risky (Pertanian → NLC §124)**
BESS Required	**No (injection < 15%)**
NEMAS Required	**Yes (>72 kWac)**
Applicant	**Solarvest Development Sdn Bhd**

Six integrated modules

1. Legal Parser — title metadata, imperial → metric, co-owner shares

2. Civil Slope Analyzer — 2D finite-difference gradients + earthworks premium

3. SLD Graph Resolver — NetworkX + TNB anti-islanding/grid-path audit

4. Financial Engine — tropical derating, degradation, BESS OPEX, IRR

5. Scenario Selector — LSS PPA / C&I NEM / self-consumption tariffs

6. Manifest Generator — SEDA & TNB prefilled forms + NEMAS checklist

Technical moat

- Junction-dot-aware SLD parsing — splits at real dots, ignores plain crossings
- Scale-invariant ε-neighbor search — works from A4 to A0 drawings
- Scenario-based finance — one model, three revenue realities
- Malaysian legal domain — NLC 1965, Ekar/Rood/Pole, SEDA NEMAS
- Offline core — symbolic solvers run locally; only OCR is cloud

Market & business model

Target customers:

- Solar EPC contractors
- Independent Power Producers (IPPs)
- Land-search / due-diligence consultancies

Go-to-market:

1. Pilot with Solarvest Innovation Lab or regional EPC — 3–5 live sites
2. B2B SaaS tiered by site-screening volume
3. Enterprise API integrations for large EPCs / IPPs

Traction & verification

[OK] 28/28 unit & integration tests passing

[OK] Streamlit app passes health checks

[OK] Public GitHub repo: github.com/lewbei/heliopolis-ai

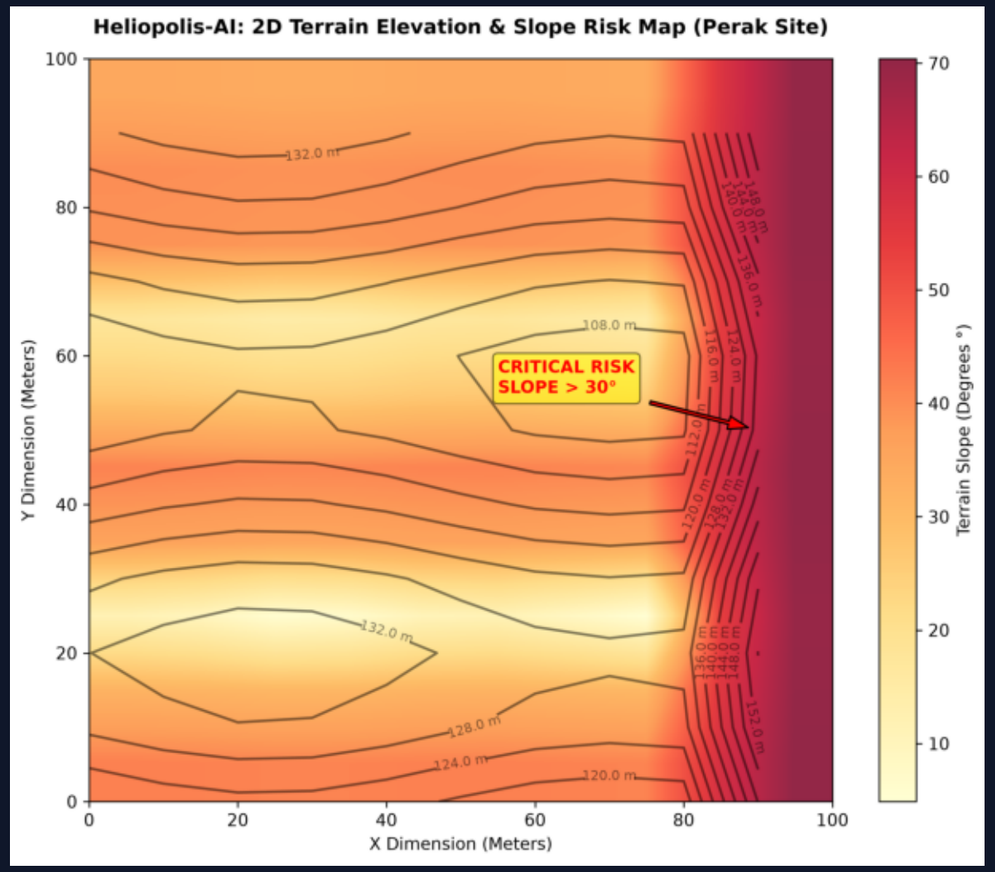
[OK] Complete QRSPI compliance folder

[OK] Positive-NPV demo output (C&I NEM: NPV RM 10.5M, IRR 12.38%)

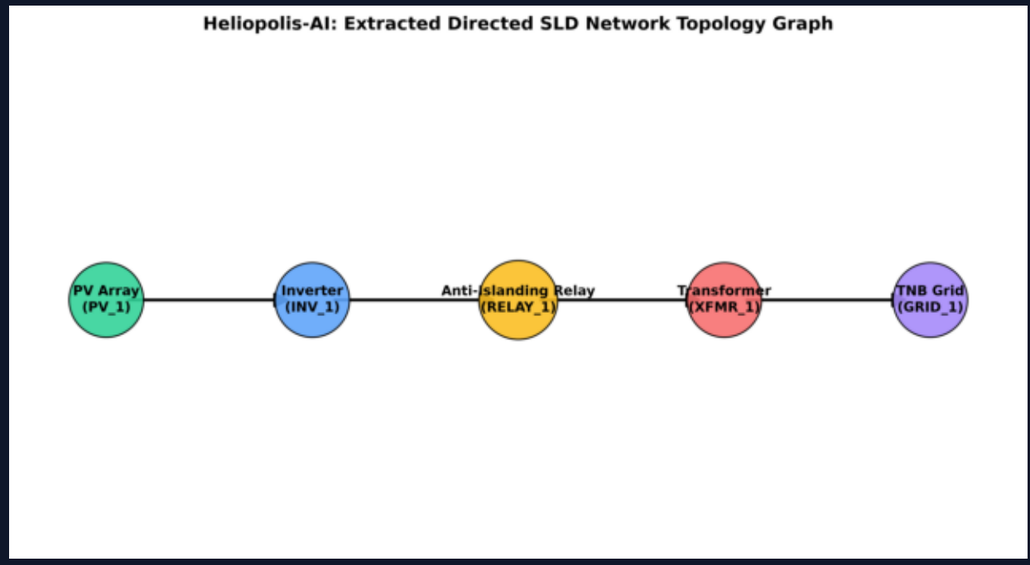
[OK] Visual outputs: slope map + SLD graph

Visual proof

Terrain Slope Risk Map



SLD Topology Flow Graph



Team

Multimedia University (MMU), Melaka, Malaysia

Software engineering · Clean-energy domain research · Malaysian regulatory knowledge

Clean-room code built on NumPy, NetworkX, Pydantic, Matplotlib, Plotly, Streamlit, Google GenAI

What we are looking for

1. Pilot partner — Solarvest Innovation Lab or a Malaysian EPC to validate on 3–5 live sites

2. Investment / prize support — to build live e-Tanah / myTNB / SEDA integrations and hosted SaaS

3. MAIC Nexus + HATI mentorship — on B2B sales and regulatory partnerships

Heliopolis-AI turns weeks of manual solar due diligence into auditable, minutes-long intelligence.

Thank you

Heliopolis-AI · Solar Project Intelligence Agent for Malaysia

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